

Arnav Singh

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UK Work Authorisation: Indefinite Leave to Remain (no sponsorship required)

PROFESSIONAL SUMMARY

MSc Data Science student at the University of Glasgow with experience developing digital solutions and process improvements across Accenture, Apollo Tyres, and Samsung R&D. Skilled at translating operational requirements into automated workflows, collaborating with technical and cross-functional teams, and using data to evaluate solutions. Demonstrated initiative through a Copilot automation proposa. Strong interest in fintech, digital banking, and creating intuitive technology-enabled customer experiences.

EDUCATION

University of Glasgow

Master of Science in Data Science

Glasgow, United Kingdom

2025 – 2026

- Dissertation: LLM-powered financial analysis and short-term stock prediction using market, filing, and news data.

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

Chennai, India

2021 – 2025

- CGPA: 9.12/10.0

SKILLS

Digital Product & Delivery: Requirements gathering, process improvement, feature delivery, Agile development, stakeholder collaboration, technical documentation

Data & Experimentation: Python, SQL, Pandas, NumPy, Microsoft Excel, data validation, exploratory analysis, model evaluation

Artificial Intelligence: Machine learning, PyTorch, scikit-learn, large language models, prompt engineering, OpenAI API, Gemini API

EXPERIENCE

Accenture

Intern

Feb 2025 – Jun 2025

- Collaborated within an Agile development team to clarify project requirements, contribute ideas, and support iterative delivery across technical workstreams.
- Identified repetitive DevOps activities and proposed a Copilot-based automation workflow to improve team efficiency and reduce manual effort.
- Worked with developers to implement REST API functionality using Java and Spring Boot, supporting the structured exchange of data across internal systems.
- Queried, reviewed, and validated MongoDB datasets used in backend operations, helping maintain accurate and reliable project information.

Apollo Tyres Global R&D

Project Intern

Jun 2024 – Sep 2024

- Gathered requirements from engineering stakeholders and translated a manual tyre-design process into an automated digital workflow.
- Developed a Python and Excel based solution supporting multiple product configurations, reducing manual effort by approximately 70%.
- Built a graphical interface that enabled engineers to select product specifications and generate AutoCAD drawings with reduced manual input.
- Validated outputs across product categories, incorporated user feedback, and added checks to improve workflow reliability and reduce opportunities for error.

EXPERIENCE (CONTINUED)

Samsung Research & Development

Feb 2023 – Sep 2023

Research Intern

- Collaborated in a four-person research team to develop an action-recognition solution using a dataset of more than 1,000 videos.
- Structured, annotated, and validated video data in JSON format, establishing reliable inputs for model development and experimentation.
- Compared model performance using quantitative evaluation metrics and communicated findings to project stakeholders.
- Contributed to the training and optimisation of VideoMAE V2 through an iterative research and evaluation process.

PROJECTS

LLM-Powered Financial Intelligence Pipeline

- Leading the end-to-end development of a financial intelligence pipeline covering 100 S&P 500 companies and combining SEC 10-K filings, financial news, market data, and LLM-generated context.
- Designed automated workflows for collecting and validating data, generating company-specific questions, running LLM inference, and producing daily financial analysis.
- Developed and evaluated predictive models for 1-, 3-, and 5-day stock-return horizons.
- Evaluated LLM configurations against answer quality, consistency, latency, and cost to support evidence-based design decisions.
- Deployed scheduled workloads using Kubernetes CronJobs and implemented duplicate-detection and recovery controls for incomplete daily outputs.

EarthFinesse: Terrain Recognition System

- Developed an end-to-end terrain-recognition application using custom image datasets, machine-learning models, and a frontend for reviewing prediction outputs.
- Prepared and validated training data, evaluated model performance, and connected the prediction workflow to a user-facing interface.

PUBLICATION

Intelligent Pipeline Inspection

- Published undergraduate research comparing deep-learning architectures for automated pipeline defect analysis using Python and PyTorch. [IEEE Xplore publication.](#)